

Two Samples, One Difference

AP Statistics · Unit 4 · Topic 4.6 · TEACHER KEY

CED TETHER

- **ENDURING UNDERSTANDING: UNC-3** — Probabilistic reasoning allows us to anticipate patterns in data.
- **Skills: 3.B** — Determine parameters for probability distributions. | **3.C** — Describe probability distributions. | **4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- **UNC-3.T** Determine parameters of a sampling distribution for a difference in sample means. [Skill 3.B]
- **UNC-3.T.1** For a numerical variable, when randomly sampling from two independent populations with means μ_1 and μ_2 and standard deviations σ_1 and σ_2 , the sampling distribution of the difference in sample means has mean $\mu = \mu_1 - \mu_2$ and standard deviation $\sigma = \sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$. **UNC-3.T.2** If sampling without replacement and sample sizes are less than 10% of population sizes, the difference is negligible.
- **UNC-3.U** Determine whether a sampling distribution of a difference in sample means can be described as approximately normal. [Skill 3.C]
- **UNC-3.U.1** The sampling distribution of the difference in sample means can be modeled with a normal distribution if both population distributions can be modeled with normal distributions. **UNC-3.U.2** If population distributions cannot be modeled with normal distributions, the sampling distribution can be approximated by a normal distribution if both sample sizes are at least 30.
- **UNC-3.V** Interpret probabilities and parameters for a sampling distribution for a difference in sample means. [Skill 4.B]
- **UNC-3.V.1** Probabilities and parameters for a sampling distribution for a difference of sample means should be interpreted using appropriate units and within the context of specific populations.

Essential question: How can two sample means use different pathways to justify one normal model for their difference?

DOK-3 skill: 4.B · **FRQ pattern:** two-shape-routes-for-a-mean-difference

DOK rationale: Part (c) coordinates a separate shape pathway for each sample mean, judges two incomplete model claims, and uses a small-sample counterfactual to trace when the tail probability loses support.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- Which distribution are we justifying for North: individual waits or sample means?
- Why can the two groups use different routes to approximate normality?
- What changes if the skewed South sample has size 8?
- What repeated-sampling event does 0.0273 describe?

[WARN] LOOK FOR

- Requiring both samples to have size at least 30 even when one population is normal.
- Assuming one normal component automatically fixes an unjustified second component.
- Subtracting variances because the statistic subtracts means.

SCORING PART (c) — E / P / I

Expected elements

- **judgment-1** (required) — Supports the original probability using North population normality and South $n=48$ CLT route.
- **judgment-2** (required) — Uses independence and explains the incomplete reasoning in both accounts.
- **judgment-3** (required) — Uses the 48-plus-8 counterexample: total 56 does not justify the small skewed South component.

E	Meets all three checks with correct contextual reasoning; equivalent wording is acceptable.
P	Shows at least one substantive correct check, but has an omission or error that prevents E.
I	Shows none of the three checks with substantive correct reasoning; a choice alone is insufficient.

[STAR] THE PROBLEM — annotated

Suppose parcel pickup waits have the shown population parameters, measured in minutes. There are 600 North visits and 1,000 South visits. North waits are approximately normal; South waits are strongly right-skewed. Independent SRSs are taken without replacement, and the service wants $P(\bar{X}_N - \bar{X}_S > 2.5)$.

Rina: “South is skewed, so I would not use a normal model for the difference.”

Oscar: “North is normal and the combined sample size is 60, so the normal model is valid; the probability is 0.0273.”

Locker samples

Locker	n	μ	σ
North	12	6.0	1.8
South	48	4.8	3.0

FIRST TAKE — one sentence before you work the parts

Would you check each place’s sample size separately or add them together? Give one reason from the study.

Key: Not graded. Each account contains a claim that still needs component-level justification.

[BOOK] JUSTIFY EACH SAMPLE MEAN BEFORE THEIR DIFFERENCE (keep on the page)

For independent random samples, $\mu_{\bar{X}_1 - \bar{X}_2} = \mu_1 - \mu_2$ and $\sigma_{\bar{X}_1 - \bar{X}_2} = \sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$ when each sample has $n \leq 0.10N$. Justify each component’s shape: a normal population supports a normal sample mean even for small n , while the Central Limit Theorem supports a sample mean from a nonnormal population when its own $n \geq 30$. The two groups need not use the same pathway. Interpret probabilities as outcomes from repeated pairs of samples. For a normal probability, standardize with $z = (\text{value} - \text{mean})/\text{SD}$ and use the appropriate normal tail. Printed normal table: $P(Z \leq 1.9220) = 0.972696$. A right-tail area is 1 minus the left area.

DOK 1 (a) State the mean of $\bar{X}_N - \bar{X}_S$ and translate the event above into context.

Key: The sampling distribution is centered at $\mu_N - \mu_S = 6.0 - 4.8 = 1.2$ minutes. The event says the sample mean North wait exceeds the sample mean South wait by more than 2.5 minutes.

DOK 2 (b) Verify both 10 percent conditions, compute the sampling-distribution SD, and calculate the probability under a normal model.

Key: North: $12 \leq 0.10(600) = 60$; South: $48 \leq 0.10(1,000) = 100$. The SD is $\sqrt{1.8^2/12 + 3.0^2/48} = \sqrt{0.4575} = 0.676387$. Thus $z = (2.5 - 1.2)/0.676387 = 1.92198$ and the upper-tail probability is $0.027304 \approx 0.0273$.

DOK 3 (c) In 4–5 sentences, decide whether 0.0273 is supported and explain what each student overlooks. Justify each sample mean using its own shape and size. Then consider $n_N = 48$ and $n_S = 8$: explain why their total alone cannot justify the same normal-model claim.

Key: The original probability is supported, but both explanations are incomplete. North’s normal population supports its sample mean at $n_N = 12$, and South’s $n_S = 48$ supports its sample mean by the CLT. Independence then supports an approximately normal difference, so Rina rejects a supported approximation. Oscar’s combined-size rule fails: with $n_N = 48$ and $n_S = 8$, the total is 56, but the small sample from skewed South has no normality justification. The total alone would make readers trust an unsupported tail approximation.